

Protocol-Guided AI Smoking Cessation Counselor: Comparing AI-Generated and Human-Curated Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as a promising approach for grounding large language model (LLM) outputs in domain-specific evidence, reducing hallucination in high-stakes clinical applications. This paper evaluates two RAG configurations for a protocol-guided AI smoking cessation counselor: (1) an *AI-generated* question-answer dataset ($n = 5,936$ pairs) and (2) a *human-curated* dataset ($n = 4,847$ pairs). We assess faithfulness, answer relevancy, context precision, and context recall using the RAGAS 0.4.x evaluation framework on a 100-question clinician-reviewed test set. Human-curated RAG achieves a mean faithfulness of 0.657 ± 0.334 , outperforming AI-generated RAG (0.531 ± 0.320) by 24% relative improvement, and reducing the high-risk hallucination rate from 43% to 34%. Compared to a web-scraping RAG baseline (WebRAG), dataset RAG is $2.41\times$ faster at cold-start and $1.44\times$ faster when web content is pre-cached, demonstrating a clear latency advantage for real-time deployment. These results underscore the clinical value of professional dataset curation and the practical benefits of vector-indexed retrieval over live web scraping in time-sensitive counseling contexts.

Keywords: Retrieval-Augmented Generation, smoking cessation, RAGAS, faithfulness, hallucination, LLM evaluation, clinical AI

1 Introduction

Smoking remains one of the leading causes of preventable death worldwide, and evidence-based cessation counseling is critical for reducing tobacco-related morbidity [who2023tobacco](#). AI-powered counseling assistants have shown promise in extending access to behavioral support, yet their clinical deployment raises important questions about factual accuracy and reliability of generated responses [thirunavukarasu2023](#).

Retrieval-Augmented Generation (RAG) addresses hallucination by grounding LLM outputs in a domain-specific knowledge base [lewis2020rag](#). However, the quality of that knowledge base—whether compiled by AI or by human experts—fundamentally determines the system’s reliability. Prior work by Ibrahim et al. established a WebRAG baseline achieving a mean faithfulness of 0.86 by retrieving live content from authoritative web sources. Our study extends

this by asking: *does the origin and curation quality of the underlying dataset matter, and at what latency cost?*

We compare three configurations evaluated against the Gemini Protocol—a structured smoking cessation framework developed by Dr. Louis Willis:

1. **Baseline LLM:** GPT-4o-mini without any retrieved context.
2. **AI-Generated RAG:** Vector retrieval from a 5,936-pair AI-generated Q&A dataset indexed in ChromaDB.
3. **Human-Curated RAG:** Vector retrieval from a 4,847-pair professionally curated Q&A dataset.

2 Related Work

Hallucination in clinical LLMs. LLMs are prone to generating plausible-sounding but factually incorrect statements, a phenomenon termed hallucination ji2023survey. In clinical settings, hallucination can mislead patients and undermine trust, making faithfulness evaluation essential singhal2023.

Retrieval-Augmented Generation. RAG lewis2020rag conditions LLM generation on retrieved documents, substantially reducing hallucination rates compared to generation-only baselines. Dense retrieval using vector databases such as ChromaDB enables efficient semantic matching at inference time.

RAGAS evaluation framework. RAGAS es2023ragas provides reference-free and reference-based metrics for evaluating RAG pipelines: faithfulness (are claims supported by context?), answer relevancy (does the answer address the question?), context precision (is retrieved context relevant?), and context recall (are reference answers covered by retrieved context?).

Smoking cessation AI. Prior work has demonstrated that LLM-based counseling assistants can deliver evidence-consistent advice eapen2023, with RAG improving adherence to clinical guidelines.

3 Methods

3.1 Test Set

We used the LFV-reviewed subset of Dr. Willis’s 150-question smoking cessation test set, excluding 23 questions flagged by a clinical reviewer, yielding $n = 127$ clean questions. We report results on the first $n = 100$ questions.

3.2 RAG Datasets

AI-Generated dataset: 5,936 question-answer pairs generated by GPT-4o-mini from the Gemini Protocol documentation.

Human-Curated dataset: 4,847 question-answer pairs professionally curated and verified against the Gemini Protocol.

Both datasets were indexed in an ephemeral ChromaDB collection using cosine similarity. Retrieval used top- $k = 3$ nearest neighbours.

3.3 WebRAG Baseline

Following Ibrahim et al., we also measure a web-scraping approach that fetches content from five authoritative sources at query time: `smokefree.gov`, CDC, Mayo Clinic, American Cancer Society, and the American Lung Association.

3.4 Answer Generation

All configurations used GPT-4o-mini (temperature = 0.3, max_tokens = 300) with a smoking cessation counselor system prompt. RAG configurations prepended retrieved context to the system prompt.

3.5 Evaluation Metrics

We report four RAGAS 0.4.x metrics:

- **Faithfulness (F):** fraction of claims in the response that are entailed by the retrieved context $\in [0, 1]$. Hallucination rate = $1 - F$.
- **Answer Relevancy:** semantic similarity between the response and a set of hypothetical questions generated from it.
- **Context Precision:** proportion of retrieved context chunks that are relevant to the ground-truth answer.
- **Context Recall:** coverage of the ground-truth answer by retrieved context.

3.6 Latency Measurement

Per-question wall-clock latency was measured using `time.perf_counter()`. Three conditions were timed: Dataset RAG (retrieval + generation), WebRAG warm (cached web content + generation), and WebRAG cold (fresh URL fetch per question + generation).

4 Results

4.1 Faithfulness and Hallucination

Table ?? presents faithfulness statistics for all configurations alongside Ibrahim et al.’s reference results.

Table 1: Faithfulness Comparison. Mean faithfulness $\in [0, 1]$, higher is better; hallucination rate $= 1 - \text{Mean}$; % High-Risk = proportion with faithfulness < 0.5 .

Configuration	Mean	SD	Median	95% CI	Halluc. Rate	% High-Risk	n
Ibrahim et al.: Base LLM	0.58	0.22	0.61	[0.42–0.74]	0.40	20%	10
Ibrahim et al.: WebRAG	0.86	0.11	0.88	[0.84–0.88]	0.14	4%	100
Ours: AI-Generated RAG	0.53	0.32	0.58	[0.47–0.59]	0.47	43%	100
Ours: Human-Curated RAG	0.66	0.33	0.70	[0.59–0.72]	0.34	34%	100

Human-curated RAG ($\bar{F} = 0.657$) outperforms AI-generated RAG ($\bar{F} = 0.531$), a difference of $\Delta = 0.126$ (24% relative improvement). Both systems exhibit substantially higher variance than Ibrahim et al.’s WebRAG ($\sigma = 0.33$ vs. 0.11), reflecting the heterogeneous nature of the static dataset compared to targeted live retrieval.

4.2 Full RAGAS Metrics

Table ?? presents all four RAGAS metrics.

Table 2: Full RAGAS Metrics (mean $\pm$ SD). N/A: metric not applicable or evaluation not completed.

Configuration	Faithfulness	Answer Relevancy	Context Precision	Context Recall
Baseline LLM	N/A	0.871 ± 0.187	N/A	N/A
AI-Generated RAG	0.531 ± 0.320	0.781 ± 0.338	0.463 ± 0.476	0.262 ± 0.330
Human-Curated RAG	0.657 ± 0.334	0.822 ± 0.289	N/A	0.395 ± 0.368

The baseline LLM achieves the highest answer relevancy (0.871), likely because its responses are unconstrained by retrieved context and optimised for question-answering fluency. Human-curated RAG shows higher answer relevancy (0.822) than AI-generated RAG (0.781), suggesting that higher-quality context also improves response quality.

4.3 Latency Comparison

Table ?? reports per-question wall-clock latency across conditions.

Table 3: Latency Comparison (seconds per question, $n = 100$). Dataset RAG uses human-curated data.

Method	Mean (s)	Median (s)	SD	p95 (s)	n
WebRAG (cold, per-Q scrape)	7.697	7.595	2.245	12.362	100
WebRAG (warm, cached)	4.620	4.275	2.571	9.572	100
Dataset RAG (human-curated)	3.198	2.866	1.605	6.316	100

Dataset RAG is $2.41\times$ faster than cold WebRAG and $1.44\times$ faster than warm (pre-cached) WebRAG. This advantage stems from ChromaDB’s sub-second vector retrieval ($\bar{t}_{\text{retrieval}} \approx 0.15\text{s}$), compared to URL fetching which averages $\approx 3\text{--}5\text{s}$ even with cached content.

5 Discussion

Faithfulness gap relative to WebRAG. Both of our RAG systems achieve lower faithfulness than Ibrahim et al.’s WebRAG (0.86). This gap reflects a fundamental architectural difference: WebRAG retrieves targeted, query-specific content from authoritative live sources, while our systems retrieve from a fixed indexed corpus. We expect that supplementing our dataset with periodic content updates from authoritative sources, or adopting a hybrid retrieval strategy (dataset + WebRAG fallback), would narrow this gap substantially.

Value of human curation. The 24% relative improvement in faithfulness from AI-generated to human-curated RAG, and the 9-percentage-point reduction in high-risk hallucination rate ($43\% \rightarrow 34\%$), quantify the clinical value of professional dataset curation. Given the sensitivity of smoking cessation counseling—where inaccurate advice may undermine quit attempts or patient trust—this quality improvement justifies the curation cost.

Latency implications. Dataset RAG’s latency advantage ($2.41\times$ vs. cold WebRAG) is practically significant for real-time chatbot applications where response times above 5–10 s degrade user experience. Even in a warm-cache scenario, Dataset RAG maintains a $1.44\times$ advantage, making it preferable for high-throughput deployment without requiring per-session web scraping infrastructure.

Limitations. (1) Context precision for human-curated RAG was not obtained due to OpenAI API rate limits during evaluation; a rerun is scheduled. (2) The test set ($n = 100$) is drawn from a single clinical protocol, limiting generalisability to other smoking cessation frameworks. (3) GPT-4o-mini was used for both answer generation and RAGAS scoring, which may introduce model-specific biases.

6 Conclusion

We present a comprehensive evaluation of AI-generated versus human-curated RAG for a protocol-guided smoking cessation counseling system. Human curation provides a meaningful improvement in faithfulness ($0.53 \rightarrow 0.66$) and reduces high-risk hallucination rates ($43\% \rightarrow 34\%$) compared to AI-generated datasets. Both systems are substantially faster than live WebRAG approaches, with Dataset RAG achieving $2.41\times$ the speed of cold WebRAG and $1.44\times$ the speed of warm WebRAG. Future work will explore hybrid retrieval (dataset + live web sources), expand the evaluation corpus beyond a single protocol, and investigate fine-tuned embeddings for domain-specific semantic matching.

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